SAM JOSEPH THIMOTHY

ROBOTICS R&D | EMBEDDED SYSTEMS | MECHANICAL DESIGN | ADVANCED MANUFACTURING

Chennai, Tamil Nadu | +91 80562 73107 | samjosephwb@gmail.com
GitHub: github.com/sam-black007 | LinkedIn: linkedin.com/in/sam-josephtimothyh007

PROFESSIONAL SUMMARY

I am a Robotics and Embedded Systems undergraduate focused on developing integrated electromechanical systems, from embedded firmware and mechanical CAD design to prototype fabrication, hardware integration and system validation. My work combines microcontroller programming, sensor and actuator integration, robotic mechanisms, motion control, UAV electronics and manufacturing workflows. Through my internships and individual engineering projects, I have worked with STM32H7, ESP32, Raspberry Pi, Arduino, IMU systems, CAN-FD, ROS-based robots and CAD-driven prototypes. I apply additive and subtractive manufacturing methods to turn design concepts into functional components, then integrate, test, troubleshoot and refine the complete system.

EDUCATION

B.E. – Computer Science, Artificial Intelligence & Robotics
Sathyabama Institute of Science & Technology, Chennai | 2024–2028

PROFESSIONAL EXPERIENCE

ROBOHATCH | Embedded Software Intern
September 2026 – Present | Chennai, Tamil Nadu | On-site
- Developing and integrating embedded systems for a seed-dropping system, contributing to controller-level functionality and hardware–software integration.
- Working with embedded devices, firmware logic and peripheral interfaces; supporting functional checks, integration and iterative debugging.

ROBOHATCH | Computer-Aided Design (CAD) Designer
August 2026 – Present | Chennai, Tamil Nadu | On-site
- Designing and refining 3D CAD components and mechanical assemblies for robotics and product prototypes.
- Revising component geometry for fit, assembly and fabrication requirements, translating concepts into prototype-ready parts.

Skylume Visions | Student Intern – Manufacturing & Prototyping
June 2026 – September 2026 | On-site
- Worked around manufacturing and prototyping workflows including CO_2 laser cutting/engraving, CNC milling, CNC routing, VMC, 3D scanning and FDM printing.
- Supported CAD-to-fabrication activities, prototype preparation, component production and mechanical fit checks.

Symbionic Pvt. Ltd. | Research & Development Intern
May 2026 – August 2026
- Supported prototype development, engineering validation and iterative testing; conducted technical literature and patent reviews.
- Collaborated with a cross-functional team to assess concepts, document findings and support prototype refinement.

Edu-versity | Machine Learning Intern
January 2026 – March 2026
- Worked on dataset preparation, preprocessing, exploratory analysis and machine-learning model evaluation using Python workflows.

SELECTED ENGINEERING PROJECTS

Collaborative team project

REX – 12-DOF Quadruped Robot | Team Development
Collaborative quadruped platform using Raspberry Pi 4/5, ROS 2 Humble in Docker, 12 RC servos and PCA9685 I²C PWM control. Project implementation includes linked-leg inverse kinematics, Bezier gait generation, PS4/PS5 teleoperation, per-servo calibration, URDF-based robot description and PyBullet simulation. My contribution should be described by the specific subsystem(s) I owned within the team. | github.com/sam-black007/zuko-robo-dog

Individual engineering projects

ARATOR – 6WD Agricultural Robotics Rover
Arduino Mega 2560-controlled field platform combining six DC drive motors with dual BTS7960 motor drivers, a soil plough, rotary earth-hogger/auger and a 4-DOF sapling-placement arm. Includes dual-joystick teleoperation, ADC deadband filtering, servo actuation and EEPROM position persistence; CAD-designed and 3D-printed mechanical assemblies. | github.com/sam-black007/arator-rover

INTENT – Gesture-Controlled Assistive Wheelchair
Dual-ESP32 wearable HMI using MPU6050 motion sensing and ESP-NOW telemetry, with gesture mapping, slew-rate PWM ramping, watchdog timeouts, emergency-stop interlock, H-bridge motor drive and attendant web console. | github.com/sam-black007/INTENT

PROMETHEUS ARM – 6-DOF Robotic Manipulator
ROS Noetic/MoveIt architecture for a 6-DOF manipulator, integrating IKFast analytical inverse kinematics, OMPL/Pilz/CHOMP/STOMP planning pipelines, Gazebo physics, ros_control PID loops, OctoMap 3D perception and safety watchdog/E-stop logic. | github.com/sam-black007/prometheus-arm

SELECTED ENGINEERING PROJECTS — CONTINUED

SYNTERRA – Precision Agriculture IoT & Plant Intelligence
ESP32 edge sensing and irrigation actuation integrated with USB serial/HTTP telemetry, Python Flask and SQLAlchemy backend, Socket.IO and React dashboard. Includes environmental/soil monitoring, DLI and microclimate calculations, irrigation watchdog logic and AI-assisted leaf pathology workflow. | github.com/sam-black007/SYNTERRA

VIGOR – 4WD Robotic Soccer Rover
4WD skid-steer mobile robot using ESP32 or Arduino firmware, dual BTS7960/L298N H-bridge drive, Bluetooth Classic SPP control, hardware PWM, directional/diagonal manoeuvres, speed steps and dynamic braking. | github.com/sam-black007/vigor

Smart AC Voltage Monitoring & Protection
Dual-controller ESP32/Arduino implementation using ZMPT101B sensing and EmonLib True-RMS computation, relay-based mains isolation, LCD telemetry, embedded HTTP dashboard, Chart.js monitoring and ESP32 NVS configuration storage. | github.com/sam-black007/smart-ac-voltage-monitoring-and-protection-system

DroneInterceptor – Drone/Bird Detection & Pi Deployment
YOLO11n two-class detection workflow with dataset acquisition, remapping, deduplication, validation, model training, ONNX export and INT8 TensorFlow Lite deployment on Raspberry Pi 5; includes target tracking and pitch/yaw/roll aim-control interface. | github.com/sam-black007/cluster-apple-pi-cam-model

HELIOS MOTION – Dual-Axis Solar Tracker | Team Contribution
Collaborative dual-MCU tracker. My documented contribution: software architecture, ESP32 IoT firmware, web-dashboard UX/UI and serial protocol; teammate handled hardware architecture, circuit implementation, actuation and Arduino UNO firmware. | github.com/sam-black007/helios-motion

TECHNICAL SKILLS

Embedded Firmware & Hardware: STM32H7, ESP32, Raspberry Pi, Arduino Mega/UNO, embedded C/C++, GPIO, ADC, PWM, timers, interrupts, peripheral drivers, firmware architecture, board bring-up, serial debugging, fault isolation and hardware–software integration

Sensors, Instrumentation & Communication: MPU6050/IMU, dual-IMU integration, LDR, DHT11, BH1750, soil-moisture and voltage sensing, sensor calibration, data acquisition, UART, I²C, SPI, CAN/CAN-FD, ESP-NOW, Bluetooth SPP, Ethernet, TCP/IP, UDP and telemetry

Robotics, Motion & Automation: ROS/ROS 2, URDF/Xacro, MoveIt, IKFast, Gazebo, PyBullet, OMPL, inverse/forward kinematics, trajectory planning, Bezier gait generation, quadruped locomotion, 4/6-DOF manipulators, servo control, DC motor drives, skid-steer platforms and joystick teleoperation

UAV & Autonomous Systems: Flight-controller integration exposure, STM32H7, IMU integration, sensor-fusion workflows, telemetry, PX4/ArduPilot exposure, multirotor integration, LiDAR-assisted mapping and computer-vision-based target tracking

Additive Manufacturing: FDM/FFF 3D printing, slicing and print preparation, build orientation, support generation, layer-based fabrication, prototype iteration, 3D scanning, dimensional/fit checks, support removal and post-processing; working knowledge of SLA/DLP and SLS principles

Subtractive Manufacturing & Machining: CO_2 laser cutting and engraving (vector/raster), CNC milling, 3-axis machining, end milling, CNC routing, PCB isolation routing, VMC, facing, profiling, pocketing, slotting, drilling, contour machining, workpiece setup awareness, toolpath/process selection and deburring/finishing

Moulding & Tooling Processes: Vacuum moulding/forming, mould and pattern preparation, forming over tooling, vacuum-assisted shaping, demoulding, trimming and finishing; prototype tooling concepts and moulded-part workflow

Advanced Manufacturing & Design Engineering: CAD/CAM workflow, design for manufacturing (DFM), design for assembly (DFA), rapid prototyping, reverse-engineering workflow using 3D scanning, process planning, material/process selection, prototype-to-part development and mechanical assembly

Software, AI & Engineering Tools: Fusion 360, SolidWorks, AutoCAD, Python, OpenCV, YOLO, TensorFlow, Flask, React, data preprocessing, model training/evaluation, Docker, Git and Linux

HANDS-ON ENGINEERING & LABORATORY EXPOSURE

- Fabrication workflow: CO_2 laser cutting/engraving, CNC milling and routing, VMC, end milling, PCB isolation routing, vacuum moulding/forming, FDM printing and 3D scanning; CAD-to-part preparation, fit checks and finishing.
- Embedded bring-up: Controller/sensor/motor-driver wiring, power and signal verification, peripheral communication checks, serial-log inspection, firmware flashing, functional tests and iterative debugging.
- Robotics integration: Servo and DC motor actuation, H-bridge drive integration, manipulator mechanisms, robot workspace/end-effector exposure, kinematic simulation and motion-control testing.
- Prototype validation: Mechanical assembly, subsystem integration, bench testing, fault diagnosis, calibration, design iteration and system-level checks.

ACHIEVEMENTS & LEADERSHIP

- Top 4 – Technathon 2025; selected for the second round of Smart India Hackathon (SIH) 2025.
- Quadruped robotics team secured 6th place in the final round of THREX Hackathon 2026.
- Supported and guided students during practical manufacturing and technical training activities.

CERTIFICATIONS & TRAINING

Simulink Fundamentals and Image Processing Onramp – MathWorks; Cloud Computing with AI – Unstop; ROBO AI – 6-week industrial training programme, My Equation; Microsoft Power BI; Python Basics and Go Programming – HackerRank; Python for Data Science – Reliance Foundation.